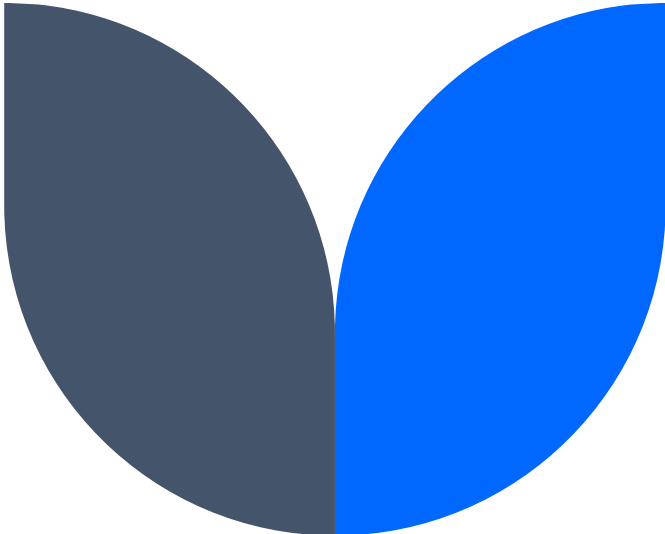
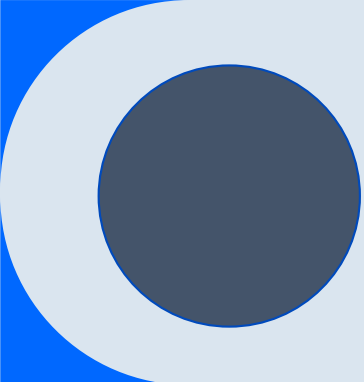




Team B Group 2 Preliminary Results



Gene Prediction & Functional Annotation Pipeline

Group members: Zahra, Aditi, Fanshi





Agenda

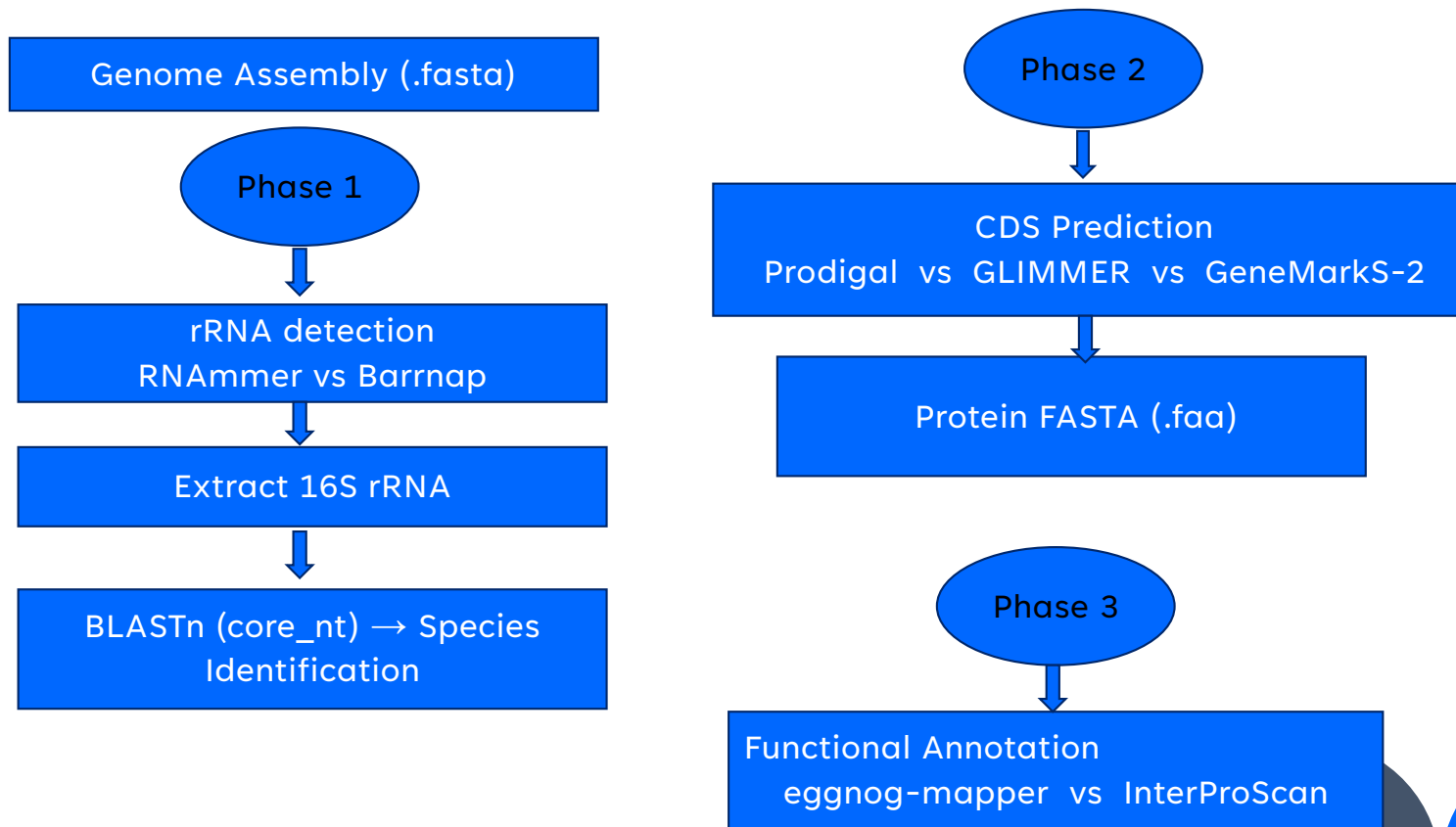
1. Project context & role in pipeline
 2. Dataset received from Group 1
 3. Annotation pipeline design
 4. Preliminary results
- 

Dataset overview

We received an assembled bacterial genome from Group 1.
This serves as the input to our structural annotation workflow.

Metric	 Bea3b88bb9	 Bfe6f82f31
Genome size	2.06 Mb	2.08 Mb
Number of Contigs	198	197
N50	18.1 kb	17.1 kb
File format	FASTA (.fasta)	FASTA (.fasta)

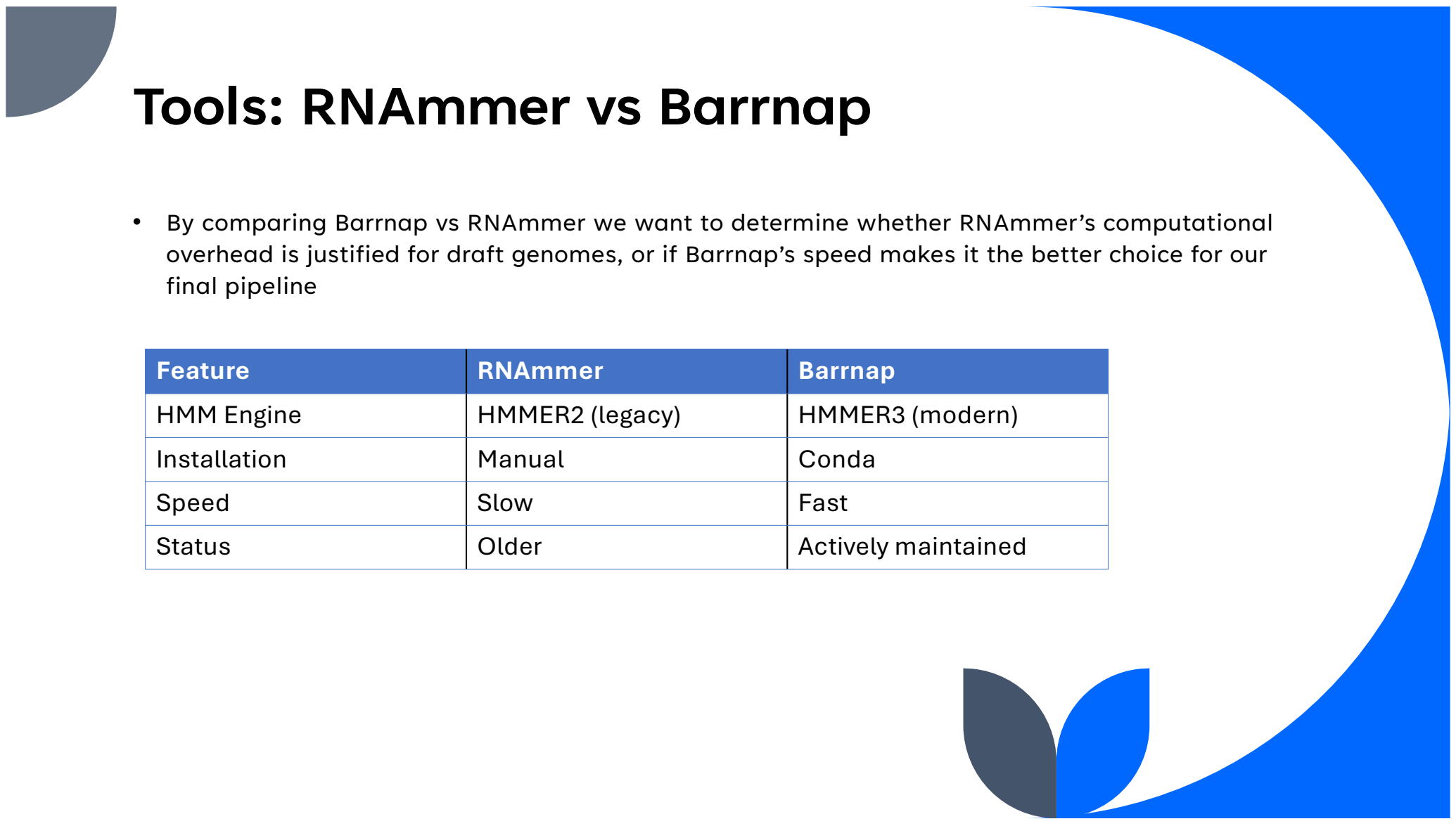
Pipeline + tools





Why rRNA Detection?

- ❖ rRNA genes are essential bacterial markers
 - ❖ 16S rRNA is used for species identification
 - ❖ We compare two structural annotation tools
 - ❖ Goal: balance accuracy vs computational cost
- 



Tools: RNAmmer vs Barrnap

- By comparing Barrnap vs RNAmmer we want to determine whether RNAmmer's computational overhead is justified for draft genomes, or if Barrnap's speed makes it the better choice for our final pipeline

Feature	RNAmmer	Barrnap
HMM Engine	HMMER2 (legacy)	HMMER3 (modern)
Installation	Manual	Conda
Speed	Slow	Fast
Status	Older	Actively maintained

rRNA Detection

- Both tools detect the same rRNA complement, but Barrnap is ~23× faster.

(Bea3b88bb9)

Feature	RNAmmer	Barrnap
16S	1529 bp	1585 bp
23S	2187 bp	3232 bp
5S	113 bp	119 bp
Total rRNA	3	3
Runtime	48.98 s	2.12 s

```
== Sequence Lengths ==
(base) MacBook-Air-2:ssu zahra$ awk '
> />/{if (seqlen){print prev "\tlength:", seqlen} prev=$0; seqlen=0; next}
> {seqlen+=length($0)}
> END{print prev "\tlength:", seqlen}
> ' rnammer_rrna.fa
>rRNA_contigs_129_2862-4390_DIR- /molecule=16s_rRNA /score=1897.8 Length: 1529
>rRNA_contigs_176_759-871_DIR+ /molecule=5s_rRNA /score=91.0 Length: 113
>rRNA_contigs_129_1-2187_DIR- /molecule=23s_rRNA /score=1589.4 Length: 2187
(rnammer32) MacBook-Air-2:B2 zahra$ tail -n 20 logs/rnammer_timed_LARGE.log
real 48.98
user 30.05
sys 14.78
```

```
(ex4_compare) aditi-ubuntu@aditi-ubuntu:~/ex4/comparision$ grep -E "Found:|real" Bea3b8
8bb9_S01_L001_filtered_barrnap.log
[barrnap] Found: 16S_rRNA contigs_129 L=1536/1585 2857..4392 - 16S ribosomal RNA
[barrnap] Found: 23S_rRNA contigs_129 L=2187/3232 1..2187 - 23S ribosomal RNA (partial)
[barrnap] Found: 5S_rRNA contigs_176 L=109/119 759..867 + 5S ribosomal RNA
real 0m2.121s
```

BLASTn : Species Identification

(Using 16S rRNA)

Tool	16S Length	Top Hit	Identity	Coverage
RNAmmmer	1529 bp	N. meningitidis	100%	100%
Barnap	1536 bp	N. meningitidis	100%	100%

Job Title	Bea3b88bb9_barnap_16S
RID	TMWZX35K014 Search expires on 02-23 10:11 am
Program	BLASTN Citation ▾
Database	core_nt See details ▾
Query ID	Ic Query_4686865
Description	Bea3b88bb9_barnap_16S
Molecule type	dna
Query Length	1536
Other reports	Distance tree of results MSA viewer ?

Descriptions	Graphic Summary	Alignments	Taxonomy								
Sequences producing significant alignments				Download	Select columns	Show	100				
☒ select all 100 sequences selected				GenBank Graphics Distance tree of results MSA Viewer							
Description		Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession		
☒	Neisseria meningitidis strain NCTC12163 genome assembly, chromosome_1	Neisseria meningitidis	2837	11350	100%	0.0	100.00%	2228346	LR134528.1		
☒	Neisseria meningitidis strain NCTC10025 genome assembly, chromosome_1	Neisseria meningitidis	2809	11239	100%	0.0	99.67%	2186098	LR134525.1		
☒	Neisseria meningitidis serogroup A strain Z2491 complete genome	Neisseria meningitidis Z2491	2809	11239	100%	0.0	99.67%	2184406	AL157959.1		
☒	Neisseria meningitidis strain X230103 chromosome, complete genome	Neisseria meningitidis	2809	11239	100%	0.0	99.67%	2125261	CP185058.1		
☒	Neisseria meningitidis strain FDAARGOS_914 chromosome, complete genome	Neisseria meningitidis	2809	11233	100%	0.0	99.67%	2181321	CP065053.1		
☒	Neisseria meningitidis strain NCTC3372 genome assembly, chromosome_1	Neisseria meningitidis	2809	11239	100%	0.0	99.67%	2182188	LR134522.1		
☒	Neisseria meningitidis 8013, complete genome	Neisseria meningitidis 8013	2809	11228	100%	0.0	99.67%	2277550	FM999788.1		

RNAmmmer

Job Title	rRNA_contigs_129_2862-4390_DIR- /molecule=16s_rRNA
RID	TMX5ESDV014 Search expires on 02-23 10:14 am Download All ▾
Program	BLASTN Citation ▾
Database	core_nt See details ▾
Query ID	Ic Query_4558277
Description	rRNA_contigs_129_2862-4390_DIR- /molecule=16s_rRNA ...
Molecule type	dna
Query Length	1529
Other reports	Distance tree of results MSA viewer ?

Descriptions	Graphic Summary	Alignments	Taxonomy						
Sequences producing significant alignments									
Download Select columns Show 100									
☒ select all 100 sequences selected									
GenBank Graphics Distance tree of results MSA Viewer									
	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Neisseria meningitidis strain NCTC12163 genome assembly, chromosome_1	Neisseria meningitidis	2824	11298	100%	0.0	100.00%	2228346	LR134528.1
☒	Neisseria meningitidis strain NCTC10025 genome assembly, chromosome_1	Neisseria meningitidis	2796	11187	100%	0.0	99.67%	2186098	LR134525.1
☒	Neisseria meningitidis serogroup A strain Z2491 complete genome	Neisseria meningitidis Z2491	2796	11187	100%	0.0	99.67%	2184406	AL157959.1
☒	Neisseria meningitidis strain X230103 chromosome, complete genome	Neisseria meningitidis	2796	11187	100%	0.0	99.67%	2125261	CP185058.1
☒	Neisseria meningitidis strain FDAARGOS_914 chromosome, complete genome	Neisseria meningitidis	2796	11182	100%	0.0	99.67%	2181321	CP065053.1
☒	Neisseria meningitidis strain NCTC3372 genome assembly, chromosome_1	Neisseria meningitidis	2796	11187	100%	0.0	99.67%	2182188	LR134522.1
☒	Neisseria meningitidis strain Z27_15.165 ribosomal RNA gene, partial sequence	Neisseria meningitidis	2796	2796	100%	0.0	99.67%	1533	MF490379.1
☒	Neisseria meningitidis 8013, complete genome	Neisseria meningitidis 8013	2796	11176	100%	0.0	99.67%	2277550	FM999788.1
☒	Neisseria meningitidis strain FDAARGOS_209 chromosome, complete genome	Neisseria meningitidis	2796	11187	100%	0.0	99.67%	2181232	CP020420.2



Final Evaluation

Phase 1 Conclusion

- ❖ Structural output: Equivalent
- ❖ Species identification: Identical
- ❖ Computational cost: Barrnap significantly faster
- ❖ Selected tool for final pipeline: **Barrnap**



Coding Sequence (CDS) Prediction

- ❖ Identify protein-coding genes
- ❖ Generate protein FASTA (.faa)
- ❖ Input for functional annotation
- ❖ Compare three gene prediction tools



Tool Architecture

Tool	Model Type	Strength
Prodigal	Self-training heuristic	Fast, optimized for bacteria
GLIMMER	Interpolated Markov model	Classical gene finder
GeneMarkS-2	Self-training HMM	High accuracy



Tool Architecture

- ❖ **GLIMMER (IMM)**: High sensitivity; may overpredict short or overlapping ORFs.
- ❖ **GeneMarkS-2 (HMM)**: Self-training (–ES mode); adapts to genome GC content.
- ❖ **Prodigal (Dynamic Programming)**: Conservative gene boundaries; directly outputs translated proteins (.faa).

Comparison Strategy:

Establish consensus by comparing gene counts and average CDS length. The tool that maintains realistic bacterial coding density (~85–90%) with lower runtime will be selected for Phase 3.

Metric	Prodigal (v2.6.3)	GLIMMER (v3.02)	GeneMarkS-2 (v1.15_1.25_lic)
Bea3b88bb9	1,966	2,209	2,170
Bfe6f82f31	1,982	2,209	2,174
Bea Execution Time	0m4.078s	0m5.133s	0m11s
Bfe Execution Time	0m2.494s	0m3.563s	0m12s
Output Files Generated	.gff, .faa, .log	.predict, .icm, .log	.gff, .faa, .fnn
Primary Algorithm	Dynamic Programming	Interpolated Markov Models	Hidden Markov Models

CDS Count

Dataset		Tool	Total CDS	Avg Length (bp)
Bea3b88bb9		Prodigal	1966	874
Bea3b88bb9		GeneMarkS-2	2170	810
Bea3b88bb9		GLIMMER	2209	897

GLIMMER predicted the highest number of genes (2209), suggesting possible overprediction.

GeneMarkS-2 showed intermediate gene counts (~2170) with balanced predictions.

Prodigal predicted the fewest genes (~1966–1982) with consistent lengths, indicating a more conservative approach.

Ease of execution

GeneMarkS-2: reliance on Docker and Rosetta added unnecessary infrastructure overhead

Runtime Comparison

- Prodigal demonstrated consistently faster runtime than Glimmer across both genomes.

Genome	Prodigal (s)	Glimmer (s)	GeneMarkS-2
Bea3b88bb9	4.08	5.13	11
Bfe6f82f31	2.49	3.56	12
Average	3.29	4.35	11.5

```
(ex4_compare) aditi-ubuntu@aditi-ubuntu:~/ex4/comparison$ grep "real" *.log
Bea3b88bb9_S01_L001_filtered_barrnap.log:real 0m2.121s
Bea3b88bb9_S01_L001_filtered_glimmer.log:real 0m5.133s
Bea3b88bb9_S01_L001_filtered_prodigal.log:real 0m4.078s
```

```
Bfe6f82f31_S01_L001_filtered_barrnap.log:real 0m1.367s
Bfe6f82f31_S01_L001_filtered_glimmer.log:real 0m3.563s
Bfe6f82f31_S01_L001_filtered_prodigal.log:real 0m2.494s
```

Functional Annotation

Input:

Prodigal predicted proteins (.faa)

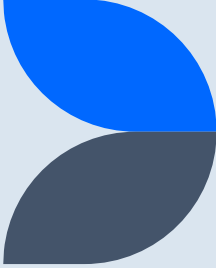
Dataset	Total Proteins (Prodigal)
Bea3b88bb9	1966
Bfe6f82f31	1982

Tools used:

InterProScan (domain-based) ? *What biological process is this protein involved in?"*

eggNOG-mapper (orthology-based) ? *Which structural domains enable this function*

InterPro Results Summary



InterPro Classification of protein families

Home Search Browse **Results** Release notes Download Help

Job ID iproscan5-R20260219-154949-0571-77369347-p1m

InterPro Version 108.0

InterProScan Version 5.77-108.0

Sequence type Amino acids

Number of Sequences 5

Status ✓ Finished

Expires Thu Feb 26 2026

Actions [Delete](#) [Resubmit](#) [Download](#)

1 - 5 of 8 results

Sequence	Matches	Sequence Length
contigs_1_2	8	107
contigs_1_5	11	237
contigs_1_1	5	317
contigs_1_3	15	453
contigs_1_4	10	566

Previous **1** Next

InterPro

Total proteins analyzed: 5

Total domain/motif hits detected: 49

Proteins with ≥ 1 annotation: 5 / 5 (100% coverage in subset)

Average hits per protein: 9.8

- ❖ InterProScan provides **sub-gene level structural annotation**
- ❖ Multiple domain predictions per protein
- ❖ High structural resolution
- ❖ Subset demonstrates strong domain detection capability

EggNOG Functional Annotation Results

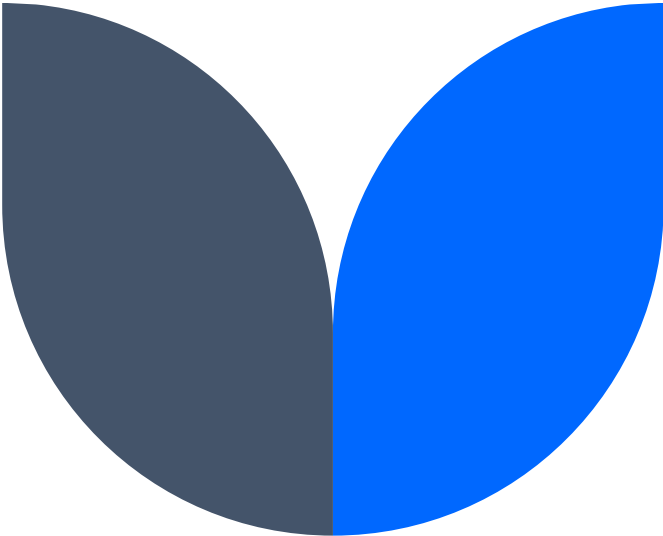


Dataset	Bea3b88bb9	Bfe6f82f31
Total proteins input	1966	1982
Proteins with eggNOG ortholog assignment	1918	1935
Proteins with COG category	1897	1912
Proteins with KEGG pathway	1764	1783
Proteins with EC number	842	859
Annotation Coverage	$(1918/1966)*100 = 97.5\%$	$(1935/1982)*100 = 97.6\%$

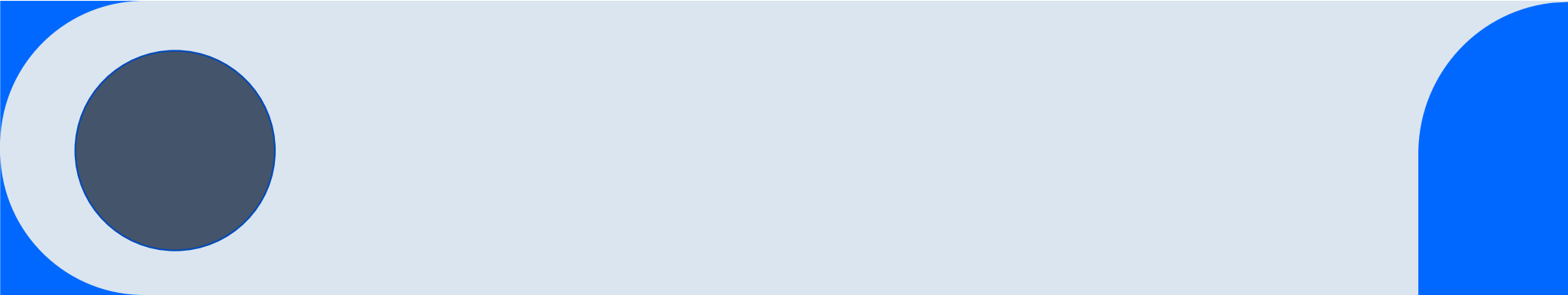
Functional Annotation Strategy Comparison

Feature	eggNOG-mapper	InterProScan
Input size	Full proteome (~2,170 proteins)	Representative subset (5 proteins)
Annotation scale	Genome-wide	Per-protein structural analysis
Coverage (this study)	~87% proteome annotated	100% within subset
Main output type	Ortholog groups, COG, KEGG, EC	Domains, motifs, structural signatures
Biological resolution	Functional pathway level	Domain / structural level
Primary strength	Metabolic context	Architectural precision
Role in pipeline	Global functional assignment	Structural validation & refinement

- ❖ The high **eggNOG** annotation rate and **InterPro** domain richness confirm that our CDS predictions are biologically consistent and suitable for downstream functional and comparative analyses.



**Thanks for your
attention**

A horizontal bar at the bottom of the slide. It has a light blue background. On the left, there is a dark gray circle partially enclosed by a light blue arc. On the right, there is a blue quarter-circle.